

LectureMind: An Algorithmic Framework for Lecture Video Understanding and Question Answering

Wang Xingxiao, Ou Zhou, Li Yibin

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1 Introduction

1.1 Background and Problem Definition

Lectures as video materials for study in universities, online courses, and professional training are quite common. In addition, they are inconvenient for efficient review at present. After returning from class, most of the time, students will no longer try to watch it all over again. Generally, there needs to be selection for definitions; determining whether an explanation of that concept is valid; making comparisons with similar concepts; asking overly specific questions. Normal video players cannot meet the needs of such uses well. They mainly support linear playback and rough time-based navigation, and cannot meet the demand of technical lectures well.

Based on this, the research question of this paper is as follows: Given a raw lecture video, how do we make it more conducive to revisiting and posing questions? We do not merely aim to produce an abstract. Instead, we combine three different concepts at a time.

- the *temporal level*, where the system identifies meaningful lecture boundaries;
- the *semantic level*, where the content is converted into explicit knowledge units;
- the *interactive level*, where users can query the lecture and obtain grounded answers.

Not every conversion will be simple in practice. Lecture audio is usually noisy; there are no synchronised slides when the content changes; the instructor spends more than a minute explaining an idea without breaking it down clearly. Due to this reason, it is necessary for the system to have reasonable multi-modal segmentation as well as a grounding-based question-answering module during lectures.

1.2 Method Overview

LectureMind, as a multiple-stage architecture for understanding the content of lecture videos. Starting with simple audio/visual analysis of speeches and texts, it then integrates these two sets of data into a relatively close form of teaching unit compared with direct time segmentation. Then each part of the whole becomes more detailed knowledge point, which is organised as summary together with mind map-style organisation in turn. On top of these representations, we support three question answering modes: direct answering without retrieval, single-pass retrieval-augmented generation, and agentic multi-step retrieval.

Figure 1 summarizes this end-to-end transformation from raw video input to interactive educational output.

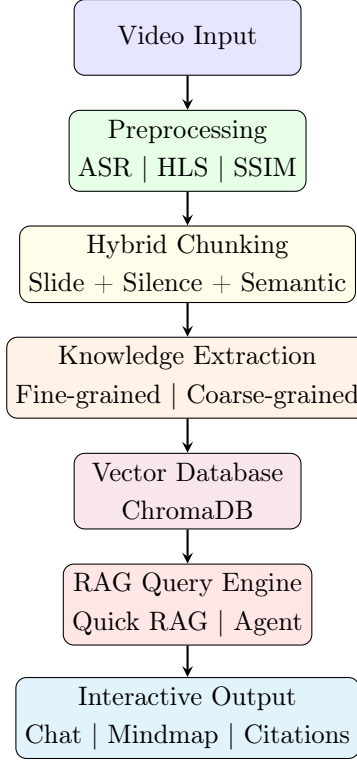


Figure 1: LectureMind System Architecture Overview

The main idea behind the framework is fairly simple: lecture understanding works better when the representation is hierarchical. Instead of treating the whole video as one long document, we break it into segments, map each segment to explicit knowledge, and organize those results into units that are small enough for retrieval but still meaningful on their own. That gives the QA system something more useful than isolated transcript sentences, while still keeping a clear link back to the original video.

1.3 Contributions

The following are the main outcomes or results achieved by this paper:

- An end-to-end approach to converting lecture videos into structured learning materials, including video Segmentation, knowledge Extraction and Grounded Question Answering.
- We propose a combination of sliding window and boundary cue strategies for chunking to enhance the stability when using only one signal.
- We propose a three-mode retrieval-aided answering architecture based on the balancing of reasoning depth, response speed and answer credibility via cascade fallback strategies.

2 Overall Framework

The LectureMind system process is divided into multiple parts. Firstly, an automatically extracted text feature is generated by the method of sound acquisition; then, a combination of image features is obtained via frame-level analysis. Neither signal is sufficient by itself. The transcript refers to what the teacher says, and the slide streaming serves as a generalisation for topic variations. We combine them to generate chunks more aligned with the transition of semantics in lectures. The above three items then form the building blocks of knowledge acquisition in the following steps.

High-level division of the pipeline at this stage includes four steps.

- **Multimodal preprocessing:** derive transcript and visual transition signals from the original video;
- **Semantic chunking:** combine visual and speech cues to partition the lecture into coherent sections;
- **Knowledge organization:** extract fine-grained knowledge points and aggregate them into higher-level structures;
- **Grounded interaction:** answer student questions using direct, retrieval-based, or agentic strategies.

To make the workflow easier to implement, the pipeline is organized as a dependency-aware task graph. Figure 2 shows the original 10-task DAG used in LectureMind for preprocessing, knowledge extraction, and summarization.

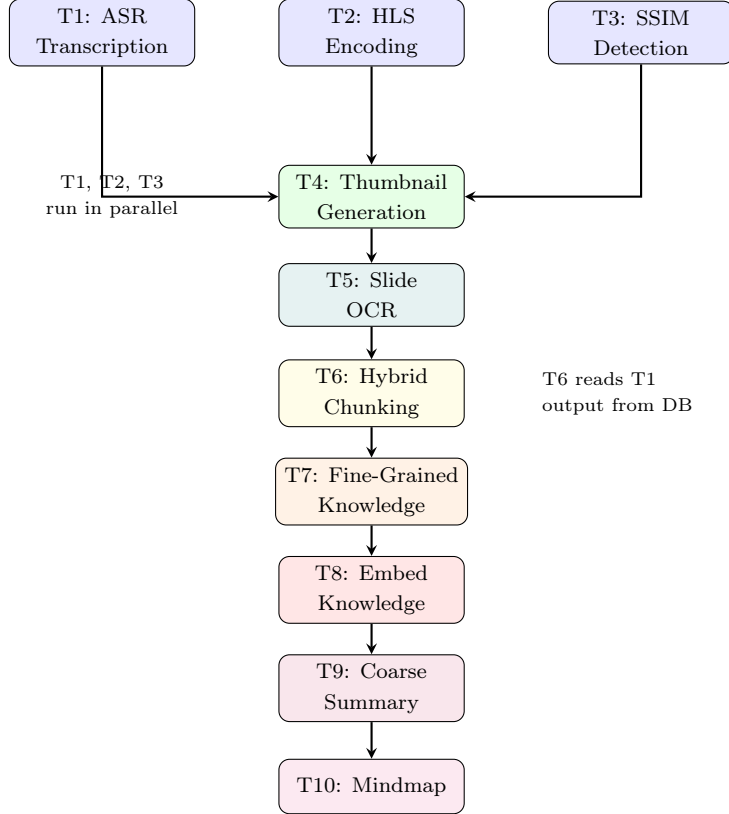


Figure 2: Task Dependency Graph (DAG) — 10 Tasks

Three design choices run through the whole framework. First, slide transition and speech time-out are considered together but are not alternatives; the lecturer’s lecture plan cannot be completely obtained from any of these modalities. Second, hierarchical knowledge representation is used. Raw transcription sentences are usually not long enough for teachers to fully grasp the lecture content; whereas an all-encompassing generalisation fails to address a particular user’s question. Third, the response strategy adapts flexibly; according to different questions, it may respond directly, retrieve data immediately, or quickly switch to an adaptive-agential mode. This is mostly an actual judgment; generally speaking, queries do not require all-optimal paths at once.

3 Lecture Preprocessing and Hybrid Chunking

3.1 Speech Transcription as a Temporal Semantic Signal

Firstly, the time-annotated text of the lecture is extracted from the speech. The above transcription has two purposes simultaneously. From a semantic perspective, it provides the core text to be employed

subsequently in knowledge retrieval and response construction processes. In terms of time, it shows where each sentence or pause is placed to represent how the teacher explains. Timestamped transcription can keep the transcript time-aligned with the video in order to answer whether a particular piece of information is supported by evidence later.

Transcripts for our task are not the final result yet; they can be seen as an intermediate form. They can determine when an example begins or ends; what is stressed at that time by the lecturer; and how a referenced answer could relate back to the lecture sequence.

Also, sentence-level timestamp data can provide assistance for the merging of close-by speech segments while preserving the time-order information. After setting the boundaries of each chunk, sentences are grouped in combination with other similar sentences to form chunks. This is because an idea may be built up over several sentences and presented clearly in a single sentence in the lecture.

3.2 Slide Transition Detection with Structural Similarity

Visual structure is captured through slide transition detection. Let I_t and I_{t+1} denote two neighboring sampled frames. Their structural similarity is measured by the SSIM function:

$$\text{SSIM}(I_t, I_{t+1}) = \frac{(2\mu_t\mu_{t+1} + c_1)(2\sigma_{t,t+1} + c_2)}{(\mu_t^2 + \mu_{t+1}^2 + c_1)(\sigma_t^2 + \sigma_{t+1}^2 + c_2)}, \quad (1)$$

where μ and σ represent local means and variances, and $\sigma_{t,t+1}$ is the cross-covariance. A substantial drop in SSIM indicates that the frame structure has changed, which is often caused by a slide transition.

This criterion works well for lecture videos because it reacts to structural changes instead of raw pixel differences. Small pointer movements or speaker motion usually do not change the overall slide layout very much, while switching to a new slide often causes a noticeable drop in structural similarity. In that sense, SSIM is more stable than naive frame differencing.

The slide signal also carries an implicit discourse prior. When an instructor advances slides, the lecture often moves to a new subtopic, example, or proof step. Of course this is not always true, but it is still a useful structural anchor, especially when the transcript alone is ambiguous.

3.3 Role of Slide Text and OCR

Some lecture content appears mainly on the slides and is only partially described in speech. Some typical examples include equations, bulleted lists, symbols, and slideshows. Therefore, we need to convert the slides into text by means of OCR or multimodal interpretation to provide an additional text channel. This decision is made to reduce the difference between what students see on the slide and what the teacher said.

To improve the extraction of this additional signal, the student may inquire about a term presented on a slide after the instructor initially mentioned it differently during lecture time. By keeping the slide text as supplementary proof, these situations become more manageable.

3.4 Hybrid Chunking

Slide transitions alone do not define all meaningful boundaries. A single slide may cover several conceptual steps, and silence-based segmentation by itself may overreact to speaking rhythm and produce fragments that are too small to be useful. LectureMind therefore combines multiple cues so that the final segments are both structurally reasonable and semantically usable. We model chunking as boundary selection over candidates derived from visual and speech cues. Let B_s denote slide-based candidates and B_p denote pause-based candidates. A candidate boundary b is scored by

$$S(b) = \alpha V(b) + \beta P(b) + \gamma C(b), \quad (2)$$

where $V(b)$ measures visual evidence from slide change, $P(b)$ measures speech evidence such as pause duration or sentence completion, and $C(b)$ optionally reflects semantic coherence between neighboring regions. In practice, obvious slide changes are treated as high-confidence boundaries, while larger speech pauses introduce extra candidates inside long slide intervals. The final set is then filtered by coherence and duration constraints. This hybrid strategy keeps the main lecture structure intact while still recovering smaller within-slide topic shifts, so the resulting chunks are short enough for retrieval but still large enough to represent a coherent teaching unit.

Table 1: Comparison of boundary cues used in chunking

Cue Type	Strength	Weakness
Slide transition	Strong structural anchor	Misses within-slide topic shifts
Speech pause	Captures local discourse rhythm	Sensitive to speaking style
Semantic coherence	Encourages topic consistency	Depends on transcript quality

4 Knowledge Extraction and Hierarchical Organization

4.1 Fine-Grained Knowledge Extraction

Following this, the lectures are divided into individual sets of explicit knowledge points. In this case, the idea is to transition from long-term explanations to smaller teaching segments that can be better stored, accessed and audited. A knowledge point can be defined as a concept explanation, method description, comparison analysis, cause-and-effect relationship elucidation, or just an essential takeaway in that part of content. Because these units are formed by semantically linked chunks rather than random transcript windows, they are often more understandable.

Fine-grained extraction has two goals: firstly, to trim the transcript to reduce management complexity; secondly, to make the pedagogical structure of the lecture clearer. Although simple keyword extraction lists keywords, this is not its purpose. Instead, we aim to achieve an intuitive understanding of what was taught in this section of the class.

Lecture materials usually consist of several parts: definitions, motivations, example problems and procedure explanation within a limited space. A good illustration needs to maintain an element of this system rather than flatten it all out.

4.2 From Chunk Content to Structured Knowledge

In each part, there is a transcript and slide shows as long as they are included. The integrated outputs of these will provide an organised account for that section. A typical knowledge point has the following five parts: a short title, a natural-language summary, representative terms, temporal links back to the lecture in intervals, and time-grounding. Time-grounding is necessary for the representation to be meaningful; when retrieving a knowledge point, there should still be an indicator pointing back to its origin in the video.

Structured knowledge points also improve retrieval granularity. If the system can only store complete-transcript content, then a specific question will likely return with considerable surrounding noise. By building indexes on smaller knowledge units, the system will be able to find targeted students' questions, including definition inquiries, simple comparisons, etc., more easily.

Table 2: Roles of different knowledge representations

Representation	Granularity	Typical Use
Knowledge point	Fine-grained	Definition and fact retrieval
Section summary	Medium-grained	Local explanation and review
Lecture summary	Coarse-grained	Global orientation
Mindmap branch	Hierarchical	Conceptual navigation

4.3 Coarse-Grained Summarization

In refining knowledge, one cannot have an all-encompassing understanding of the talk described here. All the parts of this idea are familiar to these students. Therefore, the abstracted knowledge units will be combined into coarse-grained summaries at the level of lectures or chapters. The main contents here are: key points; the order in which topics were presented; and the overall trend of the content.

Fine-grained and coarse-grained outputs do not exhibit redundancy but are hierarchically related. Fine-grained units can maintain the details for precise retrieval; coarse-grained summary provides an overall view to assist searching widely. That is to say, one level has local access while the other helps people understand the overall situation better.

4.4 Mindmap-Oriented Organization

Further organise the obtained knowledge as a tree structure, where higher-order nodes are general topics and lower-order nodes contain specific examples or supporting information. Compared to a direct listing of facts, this structure is more suitable for checking the logical organisation of a lecture. It provides an intuitive explanation for why certain parts of the lecture are closely related to some questions.

In short, this approach transforms the one-dimensional delivery of the lecture into multiple layers of knowledge resources. Video is still the source of truth; however, through the construction of a semantic layer in this system, it provides second-level support for various forms of research: immediate recall through summarization; targeting the study area by searching; and intuitively exploring the tree-like structure. This construction enables grounded and flexible answer generation in the subsequent stages.

5 Three-Mode Retrieval-Augmented Question Answering

Lecture question answering is a bit different from general open-domain QA. On the one hand, many answers depend on lecture-specific details that a base model is unlikely to know reliably. On the other hand, the questions themselves vary a lot. Some are simple factual checks, while others ask for explanations or comparisons across several lecture segments. Because of that, it is hard for a single answering strategy to work well in every case.

The following three kinds of responses are introduced by LectureMind: LLM Direct, fast RAG, and Agentic RAG. They mainly differ in how much external evidence they use and how much reasoning freedom they allow.

5.1 LLM Direct

Direct answering does not involve a specific query to the lecture knowledge base. It is suitable for general or abstract type questions and has a relatively lower response time. In addition, since there are no limitations on retrieving lecture evidence, this mode is also least suitable for conducting high-level queries of specific lectures.

5.2 Fast RAG

FastRAG performs only one retrieval operation on the indexed lecture representations when generating answers. It embeds the queries to find corresponding densely-related knowledge points, section summaries, and slide-based text in a database. Then, the most relevant items are organised into an evidence context and presented as answers. Practically speaking, this mode achieves a balance between speed and grounding at the same time. It works well for factual questions and local answers, and helps reduce the occurrence of hallucinations by using retrieved evidence. However, it remains less effective for iterative reasoning. Its main shortcoming is that it cannot deal with problems flexibly and has difficulty handling situations involving several retrievals or combining explicit evidence.

5.3 Agentic RAG

Agentic RAG extends Fast RAG by allowing iterative reasoning and tool use. Instead of relying on a single retrieval step, it can search for knowledge, inspect slide-derived text, revisit sections, and refine the evidence based on intermediate findings. This makes it more suitable for explanatory questions, cross-section comparisons, and multi-hop lecture reasoning, especially when the answer is distributed across several parts of the lecture or when the first retrieval attempt is incomplete.

Agentic RAG can be viewed as retrieval guided by explicit actions rather than a single similarity lookup. Its evidence sources may include semantic search over knowledge units, section-level details, slide-derived text, and lecture-level summaries for global orientation. Not every question requires all of these sources. The choice of tools depends on what the query is asking for.

Table 3: Typical evidence sources used by the answering system

Evidence Type	Best For	Role in Answering
Knowledge points	Local factual questions	Precise evidence grounding
Section content	Explanatory questions	Rich local context
Slide-derived text	Visual terms or notation	Recover visually expressed details
Lecture summary	Broad conceptual questions	Global context and orientation

5.4 Cascading Fallback

Although agentic reasoning is powerful, it is not always necessary and it may also introduce instability in some cases. LectureMind therefore uses a cascading fallback strategy. When appropriate, the system can start from the most capable path, but it can also fall back to a simpler mode if evidence is weak or the reasoning chain becomes unreliable. In conceptual form, the fallback chain is

$$\text{Agentic RAG} \rightarrow \text{Fast RAG} \rightarrow \text{LLM Direct}. \quad (3)$$

This design treats answering as a balance among three objectives: reasoning depth, response speed, and reliability. Agentic RAG offers the most flexibility, Fast RAG is the most balanced grounded option, and Direct mode ensures that the system can still respond when retrieval evidence is limited.

So the fallback mechanism is not only a safety feature. It is part of the answering strategy itself. The point is to use richer reasoning when it helps, without forcing every question through the most expensive and potentially brittle route.

6 Evaluation and Discussion

The evaluation focuses on three practical questions: does the hybrid preprocessing pipeline produce useful lecture segments, does the hierarchical knowledge representation support retrieval in a meaningful way, and how do the three answering modes differ in quality, robustness, and latency?

The original system evaluation was conducted on lecture videos ranging from 30 to 120 minutes and included both segmentation assessment and question answering analysis.

6.1 Processing Efficiency

Although this report is more about algorithms than deployment, processing cost still matters because the input videos are long. For 60-minute lectures, the end-to-end pipeline usually finishes in about 8 to 10 minutes when tasks overlap. Most of the time is spent in the early stages, especially speech transcription, slide detection, and visual processing. Knowledge extraction and retrieval indexing are lighter, but they are still noticeable in the overall runtime.

Table 4: Representative processing times for a 60-minute lecture

Stage	Average Time	Main Driver
ASR transcription	90 s	Audio length and quality
Slide detection	80 s	Frame sampling and SSIM analysis
Thumbnail and slide processing	80 s	Visual extraction and OCR
Hybrid chunking	5 s	Boundary fusion in memory
Knowledge extraction and summary	76 s	LLM-based semantic analysis
Indexing and mindmap generation	18 s	Embedding and aggregation

6.2 Segmentation Quality

Manual comparison against expert-annotated boundaries shows that the hybrid chunking strategy is more reliable than using only visual cues or only speech cues. On the evaluated lectures, the system achieved a precision of 0.87, a recall of 0.82, and an F1 score of 0.84. In the ablation setting, slide-only segmentation reached an F1 score of 0.76, while silence-only segmentation reached 0.71.

These results are consistent with the main assumption behind the method: lecture boundaries are better modeled by combining structural and temporal signals. Visual cues capture major transitions with relatively high confidence, while speech cues recover smaller semantic units inside long slide intervals. The fact that the hybrid method improves over both single-signal baselines suggests that the two signals are genuinely complementary in this setting.

6.3 Question Answering Performance

The three answering modes were compared using an automated evaluation setup spanning factual, conceptual, procedural, custom course-specific, and irrelevant questions. Aggregate results are summarized in Table 5.

Table 5: Aggregate comparison of answering modes

Mode	Avg. Score	Accuracy	Completeness	Hallucination	Avg. Latency
LLM Direct	69.4	66.7%	66.7%	0.0%	2866 ms
Fast RAG	71.8	71.7%	68.3%	0.0%	4767 ms
Agentic RAG	74.2	72.8%	71.1%	11.1%	5227 ms

The table shows a fairly clear pattern. Agentic RAG achieves the best average quality, which suggests that multi-step evidence gathering helps on more complex queries. Fast RAG is slightly weaker in average score, but it maintains zero hallucination in this evaluation, which highlights the value of conservative retrieval. Direct answering is still competitive on generic conceptual questions, but it is clearly less reliable when the answer depends on lecture-specific facts.

More broadly, retrieval matters most when the answer depends on lecture-specific evidence. Conceptual and procedural questions can sometimes be handled reasonably well even by direct answering. One result worth paying attention to is that stronger reasoning does not automatically mean safer answers: Agentic RAG produced the only observed hallucination, on a question where the correct behavior should have been to admit that the information was insufficient.

The evaluation also makes the latency-quality trade-off quite clear. Direct answering is fastest because it avoids retrieval altogether, Fast RAG is still reasonably responsive, and Agentic RAG is the slowest because it may perform several rounds of retrieval and reasoning before producing the final answer.

7 Conclusion and Future Work

Overall, the results suggest that the design is effective. Hybrid chunking performs better than single-signal segmentation, hierarchical knowledge representation improves targeted retrieval, and the three answering modes provide a practical balance between speed, quality, and reliability. Among them, Agentic RAG performs best on more complex questions, while Fast RAG remains the safer and more conservative option when grounded responses are the priority.

There are still some limitations. The current framework is centered on single-lecture understanding rather than course-level reasoning across many videos. The quality of knowledge extraction also depends on the quality of speech transcription and slide content. In addition, the agentic mode can become overconfident when the available evidence is incomplete, so answer calibration is still an open issue.

There are several natural directions for future work. One is cross-video retrieval, which would make the system more useful for course-level study by allowing concepts from different lectures to be retrieved and compared together. Another is more adaptive chunking that learns boundary decisions from data instead of relying mainly on heuristic fusion of multimodal signals. A third direction is richer multimodal knowledge representation, especially for equations, diagrams, and other slide content that is not fully reflected in speech. Pushing in these directions would make the system closer to a genuinely useful interactive learning assistant.

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